B TRANSITION BETWEEN INDEPENDENT VARIABLES

This appendix details the derivation of s_{tr} (short for 's transition'), the s-coordinate where the transition between the independent variables s and t occurs.

B.1 Recalled equations

The eddy viscosity is modelled using a simple closure:

$$K = \frac{\kappa u_* z}{\phi_m(z/L)}, \tag{B.1}$$

where L is the Monin–Obukhov length and ϕ_m is the Högström profile function (Högström, 1988), defined as

$$\phi_m(z/L) = \begin{cases} 1 + 5\dfrac{z}{L} & \text{for } \dfrac{z}{L} \geq 0, \\ \left(1 - 19.3\dfrac{z}{L}\right)^{-1/4} & \text{for } \dfrac{z}{L} < 0. \end{cases} \tag{B.2}$$

B.2 Derivation of the transition point

$s_{\mathrm{tr}} = t_{\mathrm{tr}}$ is defined as the specific value of s for which

$$\frac{\partial}{\partial s} = \frac{\partial}{\partial t}. \tag{B.3}$$

This criterion is chosen for convenience because it implies that the state of the system, described by the state matrix X, does not change when we switch between s and t. Note that the criterion defines a unique value because for $s = s_{\mathrm{tr}}$ we have

$$\frac{1}{\kappa} = \frac{\partial U^{0*}}{\partial t} = \frac{\partial U^{0*}}{\partial s} = \frac{1}{K^*} \tag{B.4}$$

and the equation

$$K^* = \kappa \tag{B.5}$$

has a unique solution because, regardless of stability, K^* is a monotonic function of s increasing from 0 at $s = 0$ to ∞. For $s < s_{\mathrm{tr}}$, numerical instability suggests you need to swap from s to t in the solution algorithm. To derive an equation for s_{tr}, we can start by using the chain rule to rewrite (B.3) as

$$\frac{\partial t}{\partial s} = 1. \tag{B.6}$$

Now if we substitute the definitions of $t = U^{0*}\kappa = \frac{U^0}{u_*}\kappa$ and $s = kz$, we find that

$$\frac{\partial(U^{0*}\kappa)}{\partial(kz)} = 1 \qquad \Rightarrow \qquad \frac{\kappa}{ku_*}\frac{\partial U^0}{\partial z} = 1. \tag{B.7}$$

Multiplying this through by kz and substituting the definition of ϕ_m from eq. (B.2), we can write that

$$\frac{\kappa z}{u_*}\frac{\partial U^0}{\partial z} = kz \qquad \Rightarrow \qquad \phi_m = s_{\mathrm{tr}}. \tag{B.8}$$

B.3 Stable and neutral conditions

For stable and neutral conditions ($\zeta \equiv z/L \geq 0$), there is a straightforward solution for s_{tr},

$$kz_c = 1 + 5\frac{kz_c}{kL} \qquad \Rightarrow \qquad kz_c = \frac{1}{1 - \frac{5}{kL}}. \tag{B.9}$$

B.4 Unstable conditions

For unstable conditions ($\zeta < 0$), the solution is a little more involved. We start by substituting ϕ_m and rearranging to get the quintic equation

$$\begin{aligned} kz_c &= \left(1 - 19.3\frac{kz_c}{kL}\right)^{-1/4} \\ (kz_c)^{-4} &= 1 - 19.3\frac{kz_c}{kL} \\ -\frac{19.3}{kL}(kz_c)^5 + (kz_c)^4 - 1 &= 0 \\ -\frac{19.3}{kL}s_{\mathrm{tr}}^5 + s_{\mathrm{tr}}^4 - 1 &= 0. \end{aligned} \tag{B.10}$$

This can be solved numerically using the Newton–Raphson method, which states that

$$s_{n+1} = s_n - \frac{f(s_n)}{f'(s_n)}, \tag{B.11}$$

where s_{n+1} is a better approximation of s_{tr} than s_n. This process is repeated until the relative change in s,

$$\left|\frac{\Delta s}{s_n}\right| = \left|\frac{s_{n+1} - s_n}{s_n}\right| = \left|\frac{f(s_n)}{f'(s_n)}\right|, \tag{B.12}$$

is less than a certain threshold, chosen according to the desired accuracy of the solution. In this case,

$$\begin{aligned} f(s_n) &= -\frac{19.3}{kL}s_n^5 + s_n^4 - 1 \\ f'(s_n) &= 5 \times \left(-\frac{19.3}{kL}\right)s_n^4 + 4s_n^3 \\ s_{n+1} &= s_n - \frac{-\frac{19.3}{kL}s_n^5 + s_n^4 - 1}{5 \times \left(-\frac{19.3}{kL}\right)s_n^4 + 4s_n^3}. \end{aligned} \tag{B.13}$$

We can also say something about what a suitable initial guess might be for s_{tr} depending on the value of $-\frac{19.3}{kL}$ (also known as `cdivkL` in PyFuga). If we rewrite (B.10) as

$$s_{\mathrm{tr}}^4\left(-\frac{19.3}{kL}s_{\mathrm{tr}} + 1\right) = 1 \tag{B.14}$$

then for $-\frac{19.3}{kL} \ll 1$,

$$s_{\mathrm{tr}}^4\left(-\frac{19.3}{kL}s_{\mathrm{tr}} + 1\right) \approx s_{\mathrm{tr}}^4 = 1, \quad \Rightarrow \quad s_{\text{tr, initial}} = 1. \tag{B.15}$$

Conversely, for $-\frac{19.3}{kL} \gg 1$,

$$s_{\mathrm{tr}}^4\left(-\frac{19.3}{kL}s_{\mathrm{tr}} + 1\right) \approx -\frac{19.3}{kL}s_{\mathrm{tr}}^5 = 1, \quad \Rightarrow \quad s_{\text{tr, initial}} = \left(-\frac{19.3}{kL}\right)^{-1/5}. \tag{B.16}$$

In order to cover the whole domain of values for $-\frac{19.3}{kL}$, this is implemented in PyFuga as

$$s_{\text{tr, initial}} = \begin{cases} 1 & \text{for } -\frac{19.3}{kL} > 1, \\ \left(-\frac{19.3}{kL}\right)^{-1/5} & \text{for } -\frac{19.3}{kL} \leq 1. \end{cases} \tag{B.17}$$

REFERENCES

Högström, U. (1988). Non-dimensional wind and temperature profiles in the atmospheric surface layer: A re-evaluation. *Boundary-Layer Meteorol*, 42:55–78.